

# SLUGBOT V2: An *Aplysia*-Inspired Robotic Grasper With Improved Anatomical Accuracy

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## Abstract

The sea slug *Aplysia californica* is a well-studied neuroscience model whose simple, accessible nervous system has enabled detailed studies of musculature and neural circuitry. Specifically, feeding behavior performed by its buccal mass provides a model of adaptable control and multifunctional grasping. The first version of Slug-Like Uniaxial Grasping roBOT (SLUGBOT), qualitatively mimicked the animal's feeding behavior, but lacked biologically accurate morphology. This work presents SLUGBOT V2, a second iteration of the previous *Aplysia*-inspired robot. In this version, we reduced the robot's size, updated its actuators to better match *Aplysia* morphology, and redesigned the I2 grasper protractor muscle to more accurately reflect the animal. This work demonstrated that SLUGBOT V2 more accurately reproduces key kinematics during *Aplysia* swallowing behavior, marking a significant improvement over V1.

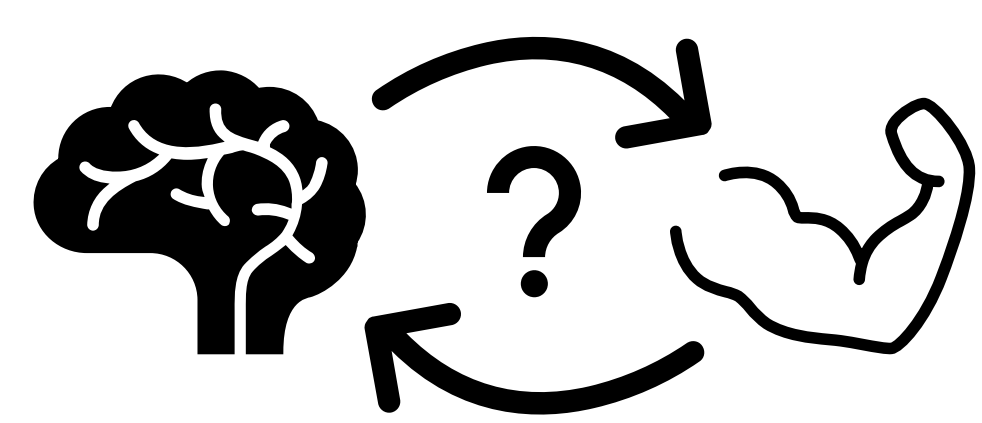
## Motivation

**We need to improve our robotic neuromechanical model to overcome the sim-to-real gap**

Much is still unknown about animal neuromuscular systems

*Aplysia californica* is a tractable model for neuromuscular control

Previous robotic model displays large sim-to-real gap



## Methods

**SLUGBOT V2 incorporated improved robot actuator design to better mimic *Aplysia* anatomy**

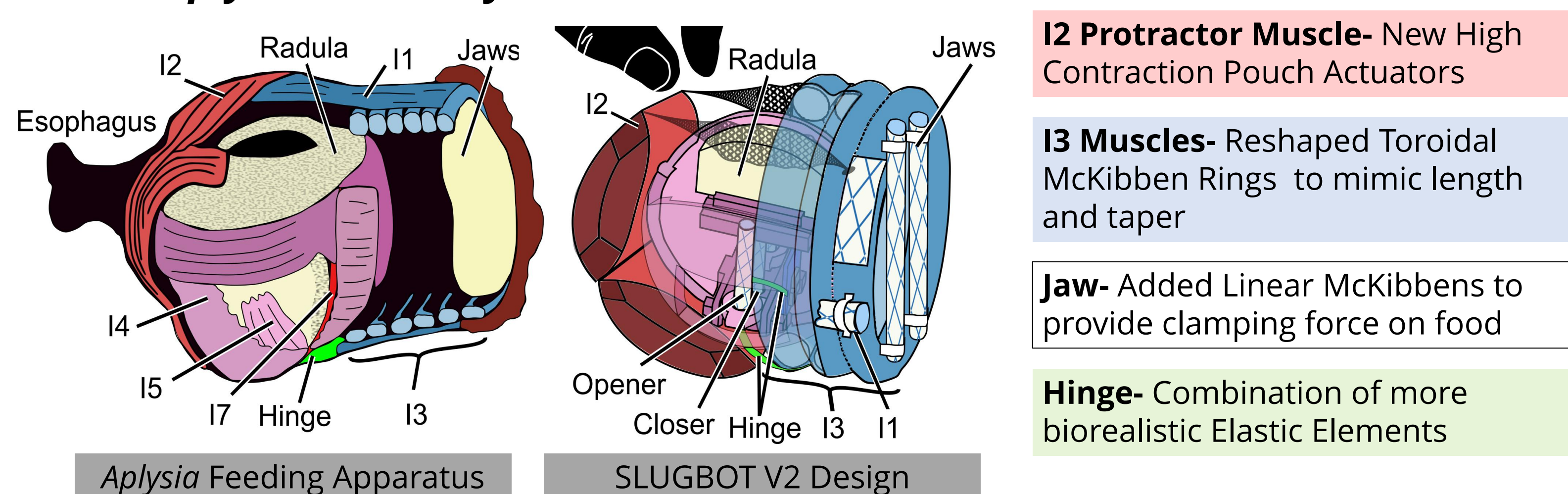


Fig. 1. *Aplysia* and SLUGBOT V2 Schematic

**Modifications made to experimental setup and robot control provide for a better comparison to *Aplysia* neural activation data during swallowing behavior**

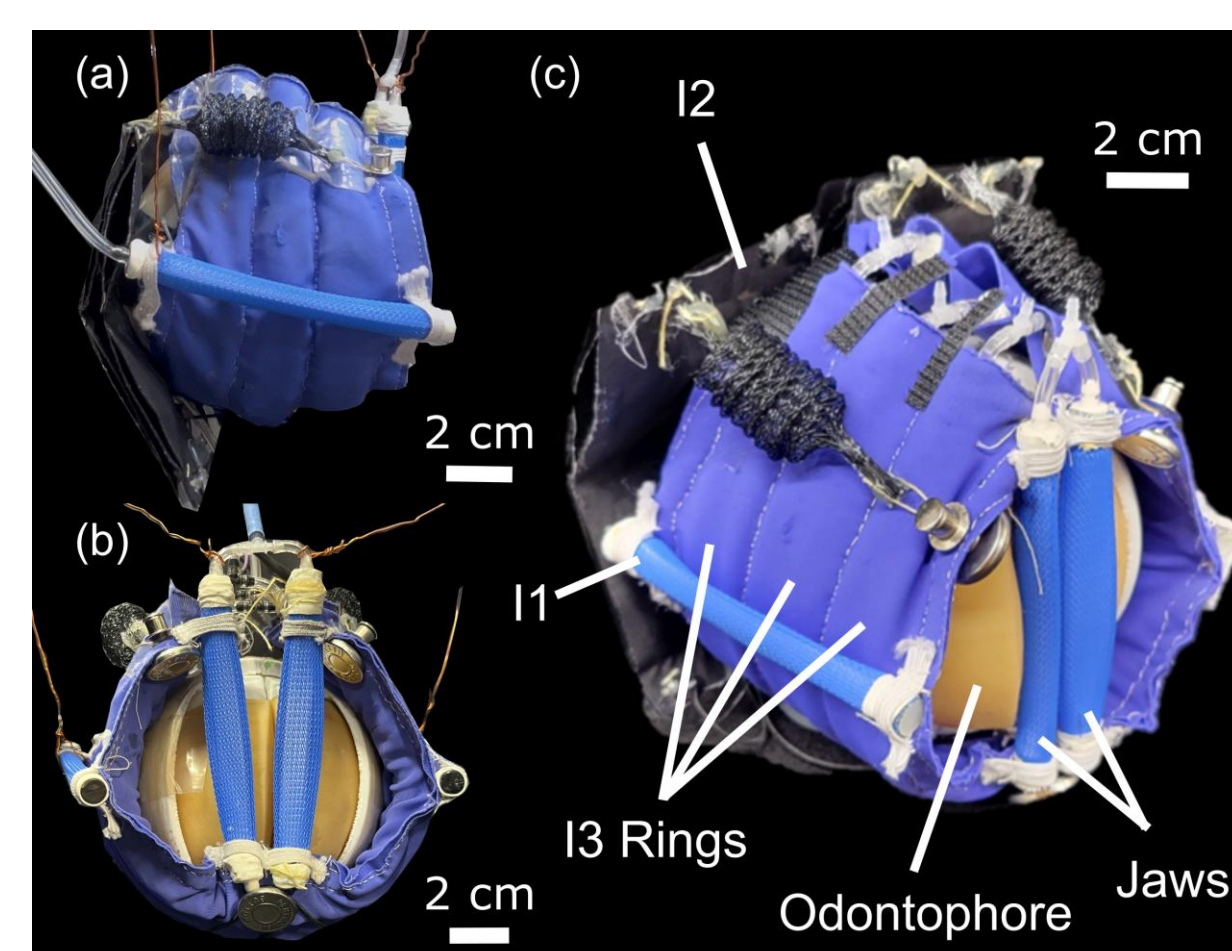


Fig. 3. a) Side, b) Front, c) Isometric Views of SLUGBOT V2

**Modified neural activation control** better captures *Aplysia* feeding behavior

**Suspension of the robot** eliminates non uniform actuator inflation

**Puppeted grasper position** is sufficiently accurate when compared to video-tracked grasper displacement

|                                  |                                          |
|----------------------------------|------------------------------------------|
| <b>Grasper Pressure</b>          | Force Sensitive Resistor (FSR)           |
| <b>Rotation</b>                  | Gyroscope (BNO 055)                      |
| <b>Puppeted Grasper Position</b> | Visually monitored using a potentiometer |
| <b>Time</b>                      | Internally tracked in system             |

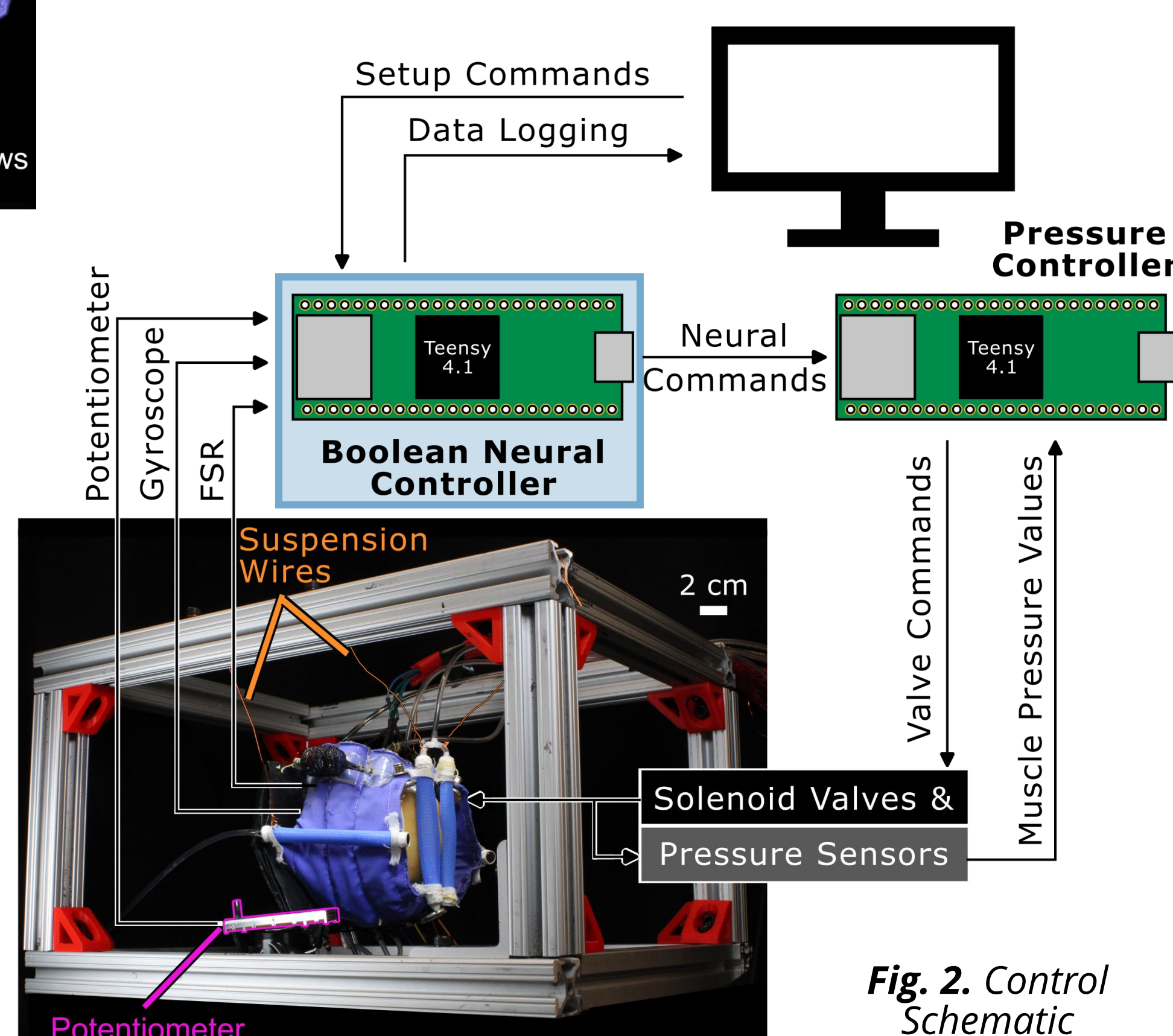
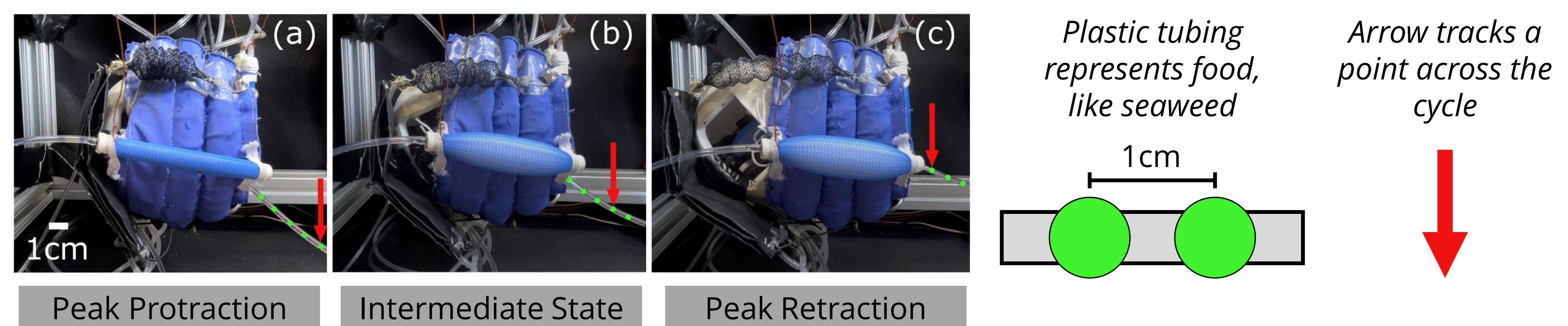


Fig. 2. Control Schematic

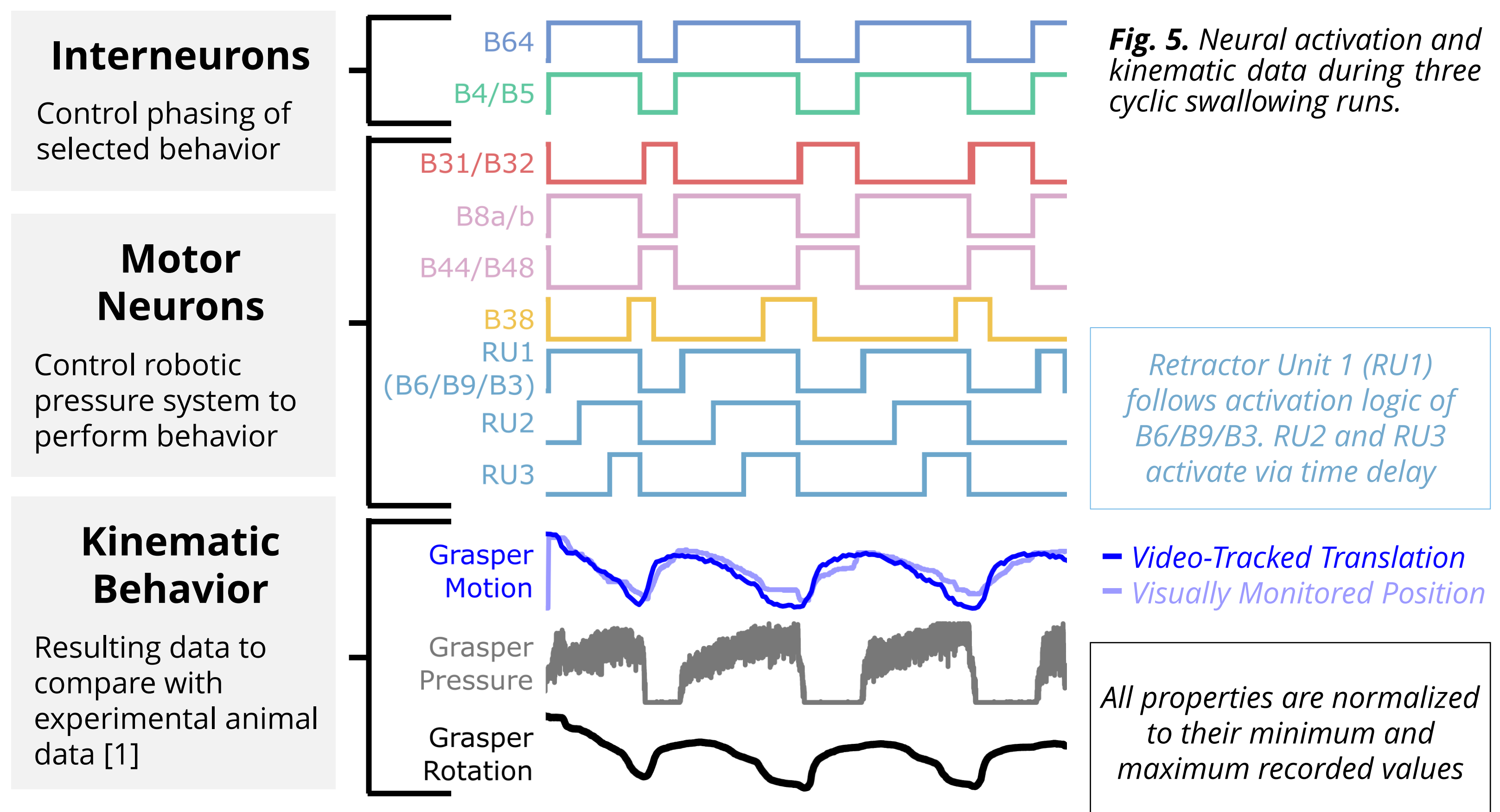
## Results: Performance Compared to *Aplysia* Data

**SLUGBOT V2 can reproduce *Aplysia*-like ingestion during swallowing**

Fig. 4. Ingestion of tube during retraction.



**Neural controller successfully produces cyclic kinematic behavior**



## Results: Performance Compared to SLUGBOT V1

**Increasing biorealism of robot model improved kinematic performance**

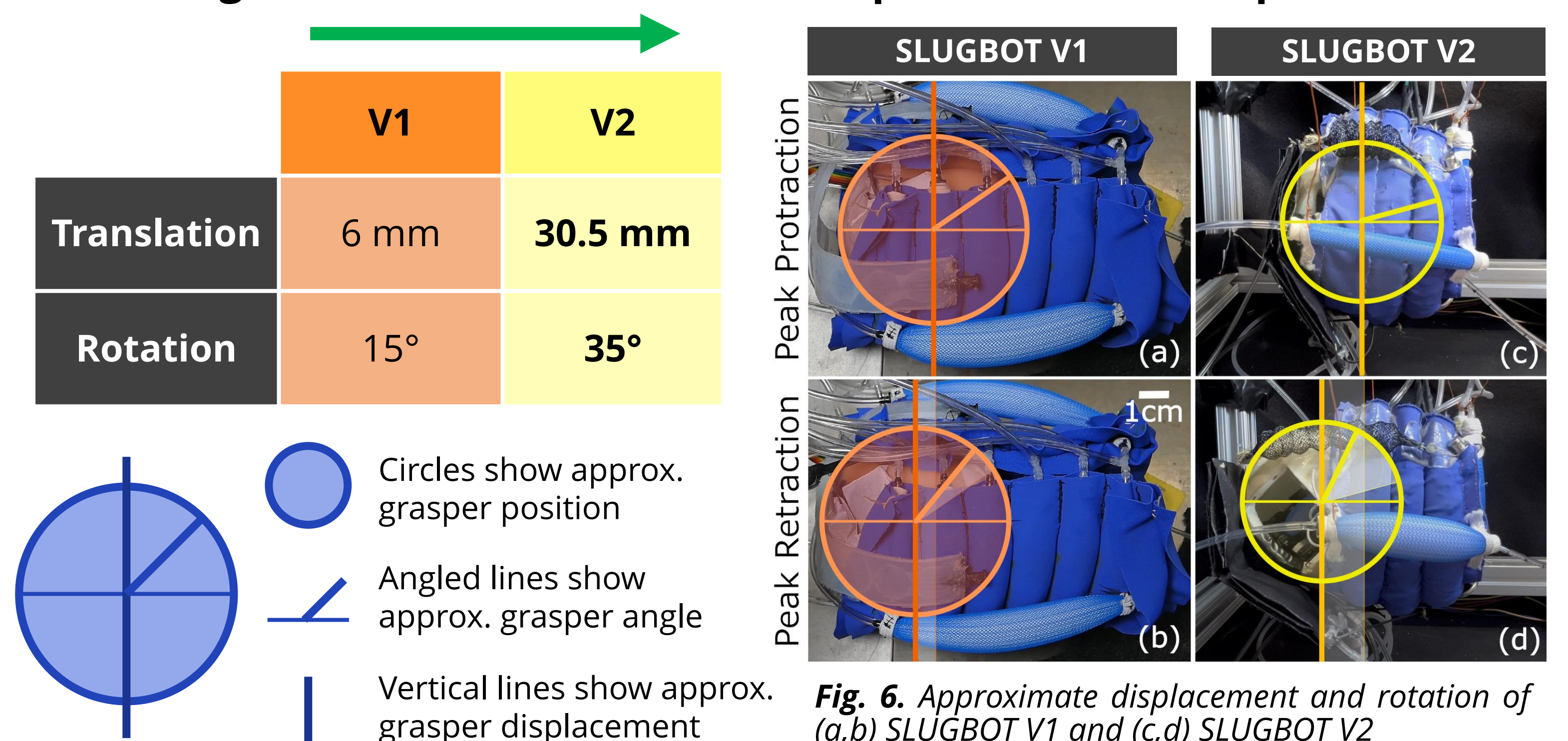


Fig. 6. Approximate displacement and rotation of (a,b) SLUGBOT V1 and (c,d) SLUGBOT V2

## Conclusions and Future Work

**Captured *Aplysia*-like translation using puppeted position control**

- Future work in making the control and sensory system fully autonomous

**Under-rotated compared to the experimental animal data**

- Incorporate grasper shape change to better mimic *Aplysia* biomechanics

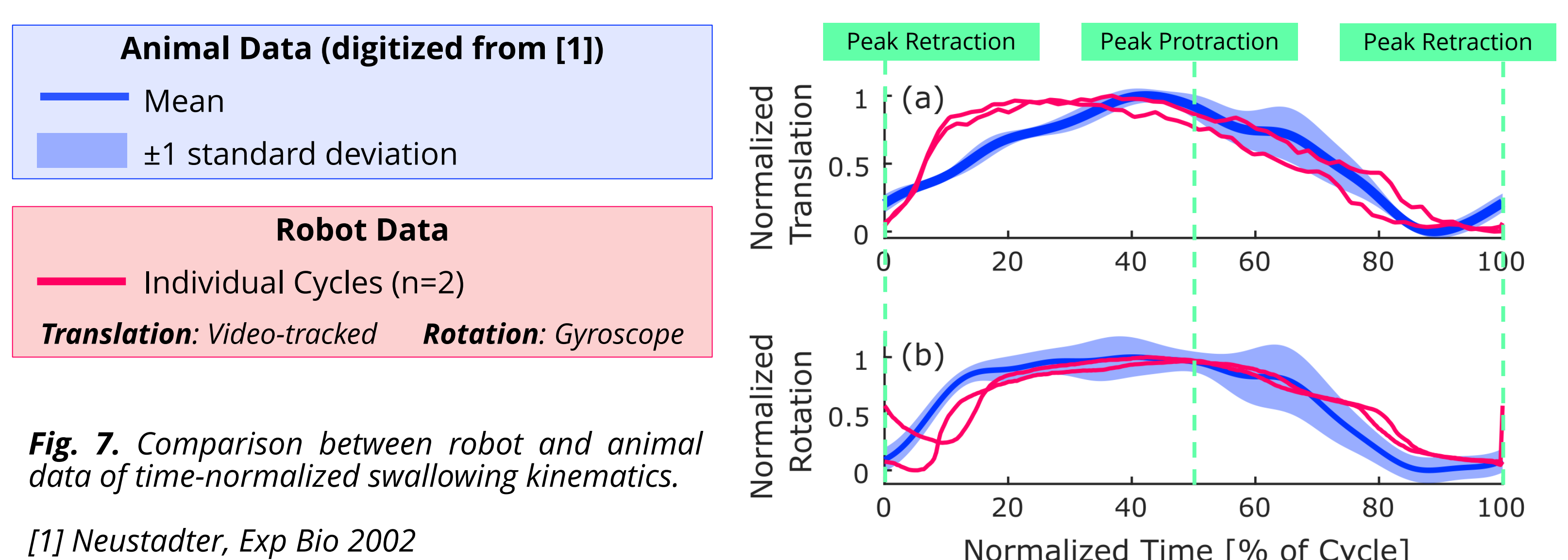


Fig. 7. Comparison between robot and animal data of time-normalized swallowing kinematics.

## Acknowledgements

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**Supplementary Videos**